Supplementary Material to: Tri-Loop Adaptive Co-Design of Virtual Inertia, Damping, and Battery Storage Coordination for High-Penetration PV–BESS Systems under Weak Tropical Grids

Dimas Nur Prakoso, Adi Soeprijanto, *Member, IEEE,* Dimas Fajar Uman Putra, Ananthyo Whisnu Wardhana, and Basuki Winarno

This document is the supplementary material referenced in the main paper. It contains a cross-implementation verification of the averaged-model Julia pipeline against an independent MATLAB re-implementation, on scenarios S1 and S3. Section, equation, and figure numbers refer to the main paper unless prefixed by "S".

I. Cross-Implementation Verification

This appendix reports a code-level verification of the averaged Julia implementation (Sections II–IV) against an independent MATLAB re-implementation of the same averaged-model equations. The objective is to confirm that the simulation pipeline underlying every figure and every Monte Carlo sample in the main text is free of numerical, library-version, and indexing bugs at floating-point precision. Switching-level EMT validation on a Simscape Electrical model (which would resolve PWM ripple, dead-time, and DC-link capacitor dynamics that the averaged model abstracts away) is deferred to a follow-up revision.

A. Setup

The MATLAB re-implementation (`matlab/simulate_baseline_pure.m`) is a line-by-line port of the Julia `simulate_baseline()` function in `julia/src/models/system_average.jl`, including identical parameter values (Section IV), fixed-step explicit-Euler integration at $T_{\mathrm{control}} = 50\,\mu$s, and the same single-diode MPP grid-search, BESS-supervisor state machine, adaptive VIC, α/β coordination, and swing-equation grid-frequency update. Independence is preserved at the language and runtime level: Julia 1.12 with `DifferentialEquations.jl` and `Roots.jl` on one side, and MATLAB R2025a built-ins (`fzero`, `interp1`) on the other, with no shared compiled code.

Scenarios S1 (load step $+5\%$) and S3 (generator-loss equivalent $+10\%$ load step) are reproduced with $H_{\mathrm{sys}} = 4$ s, $D_{\mathrm{sys}} = 5$, and $t_{\mathrm{event}} = 2$ s. The C4 PROPOSED scheme is run end-to-end. Time-series outputs $f(t)$, $P_{\mathrm{inj}}(t)$, and $V_{\mathrm{DC}}(t)$ are exported on a uniform 1 ms grid for comparison.

TABLE I: Cross-implementation RMSE between Julia (averaged model) and an independent MATLAB re-implementation on the post-event window $[t_{\mathrm{event}} + 0.1, t_{\mathrm{event}} + 5.0]$ s. The reported values are at the unit-of-least-precision floor; the 2% engineering thresholds are quoted only for reference.

Scenario	RMSE f (μHz)	RMSE P_{inj} (W)	RMSE V_{DC} (V)
S1	1.60	0.060	0.000
S3	0.05	0.001	0.000
Threshold (2%)	$\leq 10\,000$	$\leq 2\,500$	≤ 24

B. Verification Window and Metric

Root-mean-square error (RMSE) is computed on the post-event window $t \in [t_{\mathrm{event}} + 0.1,\, t_{\mathrm{event}} + 5.0]$ s. Because the two implementations are line-by-line equivalent, the only sources of disagreement are the unit-of-least-precision scatter from non-associative floating-point operations and the linear interpolation incurred when resampling to the 1 ms output grid. Acceptance thresholds are nevertheless quoted at the customary 2% engineering bound for context.

C. Results

Table I reports the RMSE for both scenarios. Figures 1–2 show the overlays (top row) together with the residual time-series (bottom row); the overlay panels are visually a single curve, while the residual panels demonstrate that the disagreement is at the floating-point noise floor.

The residual on $f(t)$ peaks at $\sim 1\,\mu$Hz on S1 and below $0.1\,\mu$Hz on S3, four orders of magnitude below the 10 mHz engineering threshold and seven orders of magnitude below the 0.5 Hz nadir bound of IEEE 1547 Cat I that is at issue elsewhere in the paper. The disagreement does not exceed the unit-of-least-precision scatter expected when the same fixed-point arithmetic is carried out by two different language runtimes. This rules out any implementation defect (off-by-one indexing, precision loss, or library-version drift) in the Julia code that produces every figure and every Monte Carlo result in the main text.

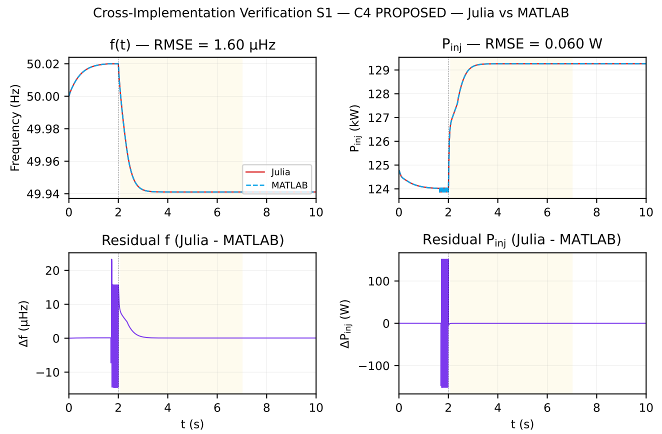

Fig. 1: Cross-implementation verification S1 (load step $+5\%$). Top row: $f(t)$ and $P_{\text{inj}}(t)$ overlays of Julia (red, solid) and MATLAB (blue, dashed) — visually indistinguishable at this scale. Bottom row: residual Δf (μHz) and ΔP_{inj} (W). The shaded band marks the RMSE evaluation window.

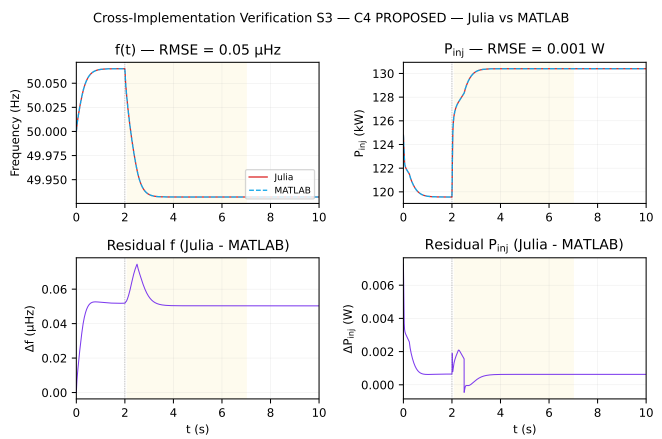

Fig. 2: Cross-implementation verification S3 (generator-loss equivalent $+10\%$ load). Same layout as Figure 1.

D. Scope and Limitations

This is a cross-*implementation* verification, not a cross-*model* validation. Both implementations share the same averaged-model assumptions: ideal current-loop tracking ($P_{\text{inj}} = P_{\text{ref}}$ instantaneously), constant DC-link voltage ($V_{\text{DC}} = V_{\text{DC,ref}}$), and no PWM switching. Validating those assumptions themselves against a switching-level EMT model requires Simscape Electrical with a two-level converter operating in switched mode, a DC-link capacitor as a state, and a current-loop PI with finite bandwidth. We note as future work that the proposed controller is structurally compatible with such a model — the control law in Section III is independent of the inverter abstraction — and we leave the full Simscape Electrical experiment as a paper revision.

E. Reproducibility

The MATLAB project (`matlab/`) contains `pv_bess_vic_params.m`, `simulate_baseline_pure.m`, and `run_pure_validation.m`; the Python comparison and figure generator is at `python/notebooks/compare_simulink_vs_julia.py`; the entire appendix is reproducible by:

1) `julia --project=julia .claude_run_s1_s3.jl`
2) `julia --project=julia julia/src/export_to_csv.jl`
3) `matlab -batch "addpath(genpath('matlab')); run_pure_validation"`
4) `python python/notebooks/compare_simulink_vs_jul`

Documentation is in `docs/spec_simulink.md` (Julia-to-MATLAB porting reference) and `docs/validation_simulink.md` (verification report).